```
      IF (LAG1 .NE. LAG + 1) IFAULT = 13
      NB = 0
      IF (IFAULT .NE. 0) RETURN
C
      IQ = 0
      DO 300 M = 1, LAG
      IP = LAG + 1 - M
      J = IQ + 1
      DO 100 L = J, M
      IQ = M + J - L
C
C        IF TABLE DOES NOT CONTAIN PROBABILITIES OF SIGNIFICANCE
C        BUT T OR CHI2 STATISTICS, REPLACE .LT. BY .GT. IN :
C
      IF (ABS(TABLE(IP, IQ)) .LT. CRIT) GOTO 300
  100 CONTINUE
  200 IQ = J - 1
  300 MINP(IP) = IQ
      MINP(LAG1) = 0
      IF (MINP(1) .EQ. 0) GOTO 500
      INDP = LAG1
      DO 400 M = 1, LAG1
      IF (MINP(M) .EQ. INDP) GOTO 400
      NB = NB + 1
      MDLP(NB) = M - 1
      MDLQ(NB) = MINP(M)
      INDP = MINP(M)
  400 CONTINUE
  500 RETURN
      END
```

Remark AS R73

A Remark on Algorithm AS 222: Resistant Smoothing using the Fast Fourier Transform

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There are two errors in the algorithm given by Härdle (1987).

First, if subroutine *RESSMO* is called with its *NEWCAL* parameter equal to unity, use is made of the values in rows 5 and 6 of the *WK* array in the loop immediately before statement 56. These rows have never been assigned values, as the algorithm stands. The following statements should be inserted immediately after the call to *LINSMO* which follows statement 56:

```
      DO 65 J = 1, NOUT
      WK(5, J) = WA(J)
65    WK(6, J) = WB(J)
```

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Secondly, in the auxiliary subroutine *LINSMO*, the statement

$$M1 = M/2$$

should be moved from immediately after statement 20 to become the first executable statement of the subroutine. The value of variable $M1$ is used on all entries to *LINSMO*, in the call to *FASTF* which appears immediately before the end of the subroutine. At present, however, $M1$ is assigned a value only if *LINSMO* is called with its *NEWCAL* parameter equal to zero. Unless the compiler happens to preserve the values of local variables between successive calls to *LINSMO*, this will result in an inappropriate value for $M1$ in the call to *FASTF*. Algorithm AS 222 uses the original version of the *FASTF* routine (Monro, 1975), which does not check this parameter, unlike the revised version (Griffiths and Hill, 1985).

References

Griffiths, P. and Hill, I. D. (eds) (1985) *Applied Statistics Algorithms*. Chichester: Horwood.
Härdle, W. (1987) Resistant smoothing using the fast Fourier transform. *Appl. Statist.*, **36**, 104–111.
Monro, D. M. (1975) Complex discrete fast Fourier transform. *Appl. Statist.*, **24**, 153–160.

Remark AS R74

A Least Squares Version of Algorithm AS 211 : The *F–G* Diagonalization Algorithm

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Introduction

The *F–G* diagonalisation algorithm of Flury and Constantine (1985) finds the 'common principal components' of $k \geqslant 1$ positive definite symmetric $p \times p$ matrices A_i, $i = 1, \ldots, k$. The goal is to find an orthogonal matrix B such that $B'A_iB$ is nearly orthogonal for all i. Algorithm AS 211 computes maximum likelihood estimates of B assuming that each matrix A_i is distributed according to a Wishart distribution. De Leeuw and Pruzansky (1977) have given a least squares algorithm for computing common principal components in the context of multidimensional scaling. An advantage of the least squares criterion is that the matrices A_i are not required to be positive definite. The purpose of this remark is to give a modification to Algorithm AS 211 which implements the De Leeuw and Pruzansky (1977) algorithm.

Least Squares Version of AS 211

The criterion used in the least squares algorithm is for symmetric matrices only

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